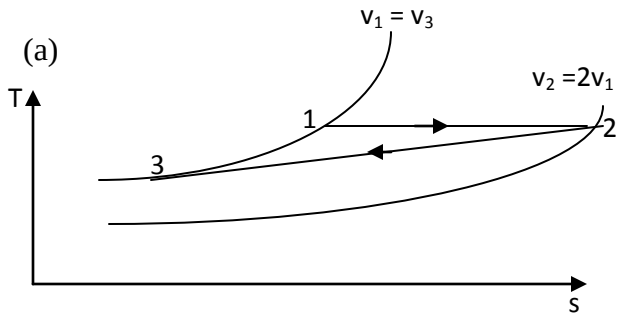
SECTION B

QUESTION 1 (20 points)

SOLUTION

(a)



(b)

$$T_1 = T_2;\ \frac{v_2}{v_1} = 2;\ P_2 = P_3;\ v_3 = v_1 \Big] \text{ Given}$$

For an ideal gas

$$P\,v = R\,T,$$

Hence for state 2, we write $P_2 v_2 = RT_2$.....(i)

And for state 3, we write $P_3 v_3 = RT_3$.....(ii)

Equating (i) and (ii) therefore we have

$$\frac{P_3 v_3}{P_2 v_2} = \frac{T_3}{T_2} \Rightarrow \frac{v_3}{v_2} = \frac{T_3}{T_2} \Rightarrow \frac{v_1}{2v_1} = \frac{T_3}{T_1}$$

$$\text{Thus } T_3 = \frac{T_1}{2} = \frac{533\,\text{K}}{2} = 266.5\,\text{K}$$

(c)

For entropy change of an ideal gas, we can write either of these two equations for a process starting at state 'i', initial state and ending at state 'o', final state.

$$\Delta S = S_o - S_i = m\left(c_v \ln\left(\frac{T_0}{T_i}\right) + R \ln\left(\frac{v_o}{v_i}\right)\right).(1)$$

$$\Delta S = S_o - S_i = m\left(c_p \ln\left(\frac{T_0}{T_i}\right) - R \ln\left(\frac{P_o}{P_i}\right)\right)...(2)$$

However, to determine the entropy change of process 1 to 2, it is appropriate to use equation (1) since it is an isothermal process and the volumes are also provided in the question. Hence we write

0 since $T_1 = T_2$

$$(\Delta S)_{12} = S_2 - S_1 = m\left(c_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{v_2}{v_1}\right)\right)$$

Therefore,

$$(\Delta S)_{12} = S_2 - S_1 = m\,R \ln\left(\frac{v_2}{v_1}\right) = 1\,\text{kg} \times 0.287\,\frac{\text{kJ}}{\text{kg.K}} \ln(2).$$

$$= 0.199\,\text{kJ/K}$$

Also, to determine the entropy change of process 2 to 3, it is appropriate to use equation (2) since it is an isobaric (constant pressure) process and the temperatures are known. Hence we write

0 since $P_2 = P_3$

$$(\Delta S)_{23} = S_3 - S_2 = m\left(c_p \ln\left(\frac{T_3}{T_2}\right) - R \ln\left(\frac{P_3}{P_2}\right)\right)$$

Therefore,

$$(\Delta S)_{23} = S_3 - S_2 = -m\,c_p \ln\left(\frac{T_3}{T_2}\right)$$

$$= -1\,\text{kg} \times 1.005\,\frac{\text{kJ}}{\text{kg.K}} \ln\left(\frac{266.5}{533}\right) = \text{-0.696 kJ/K}$$

Hence the overall entropy change (net entropy change) is

$$(\Delta S)_{net} = (\Delta S)_{12} + (\Delta S)_{23} = +0.199 - 0.696$$
$$= -0.497 \text{ kJ/K}$$

(d)

For process 1 to 2, we have a reversible isothermal process and hence the heat supplied during the process is given by

$$Q_{12} = T_1 \, dS = T_1 (\Delta S)_{12} = 533 \text{ K} \times 0.199 \text{ kJ/K}$$
$$= 106 \text{ kJ}$$

(e)

For process 2 to 3, we have a reversible isobaric (constant pressure) process and hence the heat rejected during the process is given by

$$Q_{23} = (\Delta H)_{23} = m(h_3 - h_2) = m c_p (T_3 - T_2)$$
$$= 1 \text{ kg} \times 1.005 \frac{\text{kJ}}{\text{kg. K}} \times (533 - 266.5) \text{K}$$
$$= -268 \text{ kJ}$$

(f)

Hence, the overall heat transfer during the two processes (net heat transfer) is given by

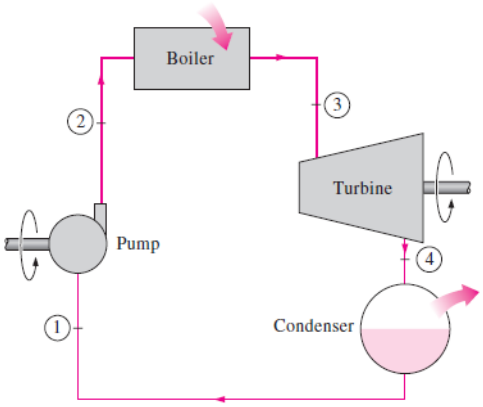
$$Q_{net} = Q_{12} + Q_{23} = 106.1 - 268 = -161.9 \text{ kJ}$$

The negative sign indicates that the net flow is heat rejection.

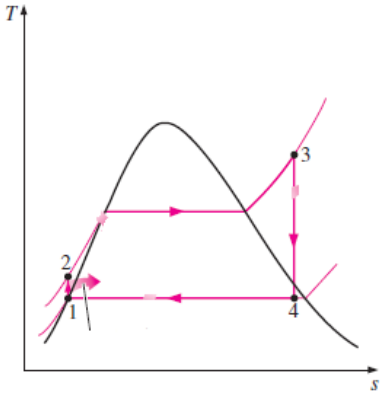
QUESTION 2 (20 points)

SOLUTION

Flow diagram of the problem



Corresponding T-s diagram



(a)

To determine the specific enthalpies of the working fluid at each of the four state points, thus h_1, h_2, h_3, and h_4, we make use of the values of steam properties provided in Table 1.0 on page 12-15 of 17 attached.

State 1

The working fluid is exactly saturated liquid and hence the specific enthalpy at state 1, h_1 will correspond to the h_f value at that pressure $P_1 = 0.065$ bar located on page 12 of 17.

Thus, $h_1 = h_{f\,@\,0.065\text{bar}} = 158\,\text{kJ/kg}$

State 2

To determine the work input to the feed pump, we write

$W_{12} = h_2 - h_1$, but since $W_{12} \approx 0$
It means $h_2 \approx h_1 = 158\,\text{kJ/kg}$

State 3

The working fluid is in the superheated state as indicated on the T-s diagram, hence the specific enthalpy and specific entropy at state 3, h_3 and s_3 will correspond to the 'h' and 's' values at that pressure P_3 = 40 bar and temperature T_3 = 350 °C located on page 12 of 17.

$$\left.\begin{aligned} P_3 &= 40\text{bar} \\ T_3 &= 350\,°\text{C} \end{aligned}\right\} \text{Superheated steam}$$

$h_3 = h_{@\,P=40\text{bar},t=350°\text{C}} = 3094\,\text{kJ/kg}$

$s_3 = s_{@\,P=40\text{bar},t=350°\text{C}} = 6.584\,\text{kJ/kg.K}$

State 4

The turbine process 3-4 is an isentropic process as indicated on the T-s diagram and hence it indicates that the specific entropy at state 3, s_3 is the same as the specific entropy at state 4, s_4.

Thus, $s_4 = s_3 = 6.584\,\text{kJ/kg.K}$

The pressure at state 4 is 0.065 bar and the specific entropy of the saturated vapour at this pressure, s_g is 8.301 kJ/kg.K.

Since s_4 is less than s_g at 0.065 bar, it implies that state 4 is a saturated liquid vapour mixture state. Hence the quality of steam at state 4 is determined as;

$$x_4 = \frac{s_4 - s_{f\,@\,0.065\text{bar}}}{s_{fg\,@\,0.065\text{bar}}} = \frac{6.584 - 0.541}{7.76}$$
$$= 0.779$$

Similarly, the specific enthalpy at state 4 is determined as;

$$h_4 = h_{f\,@\,0.065\text{bar}} + x_4 h_{fg\,@\,0.065\text{bar}}$$
$$= 158 + 0.779\,(2412) = 2037\,\text{kJ/kg}$$

(c)

Since the feed pump work is negelected, the net work output per unit mass of steam is equal to the turbine work output. Thus;

$W_{net} = W_{34} + W_{12} = W_{34}$ since $W_{12} \approx 0$
$W_{net} = W_{34} = h_4 - h_3 = 2037 - 3094$
$= -1057\,\text{kJ/kg}$
$|W_{net}| = 1057\,\text{kJ/kg}$

(d)

$$\eta_{th} = \frac{|W_{net}|}{Q_{23}} = \frac{h_3 - h_4}{h_3 - h_2} = \frac{3094 - 2037}{3094 - 158} = 0.36$$